TSA Claims Case Study (2002–2017)

Tool Used: SAS OnDemand for Academics

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Objective

This case study explores over 220,000 TSA claims filed between 2002 and 2017. The goal is to uncover insights into trends by year, claim types, payout patterns, and geographic hotspots to inform operational improvements and risk mitigation strategies.

Tools Used

- SAS Studio (OnDemand)
- PROC IMPORT, PROC FREQ, PROC MEANS, PROC GCHART, and more
- Visual outputs exported as PNGs

Data Overview

Rows: 220,855
Columns: 14
Source: TSAClaims2002_2017.csv
Key Variables: Incident Date, Claim Type, Claim Site, Close Amount, State, Disposition

Data Cleaning

Removed rows with missing Incident_Date
Filtered out invalid or outlier dates (pre-2000)
Standardized claim type labels
Verified numeric consistency in payout fields

Key Insights

1. TSA Claim Volume by Year

TSA claims peaked in the early 2000s and gradually declined, possibly reflecting improved screening protocols.

The FREQ Procedure

Year	Frequency	Percent	Cumulative Frequency	Cumulative Percent
1996	1	0.00	1	0.00
2000	14	0.01	15	0.01
2001	15	0.01	30	0.01
2002	2158	0.99	2188	1.00
2003	24508	11.22	26696	12.22
2004	28960	13.25	55656	25.47
2005	22971	10.51	78627	35.98
2006	18806	8.61	97433	44.59
2007	18275	8.36	115708	52.95
2008	15902	7.28	131610	60.23
2009	11179	5.12	142789	65.34
2010	12215	5.59	155004	70.93
2011	10995	5.03	165999	75.96
2012	9997	4.57	175996	80.54
2013	9536	4.36	185532	84.90
2014	8680	3.97	194212	88.87
2015	7721	3.53	201933	92.41
2016	8188	3.75	210121	96.15
2017	8403	3.85	218524	100.00
2018	2	0.00	218526	100.00
Frequency Missing = 2329				

2. Most Common Claim Types

The majority of claims were related to property damage and loss, with fewer injury claims reported.

The FREQ Procedure

Claim_Type	Frequency	Percent
Passenger Property Loss	126795	57.41
Property Damage	82603	37.40
Unknown	7908	3.58
Personal Injury	1613	0.73
Employee Loss (Mpceca)	492	0.22
Passenger Theft	479	0.22
Motor Vehicle	423	0.19
-	375	0.17
Complaint	73	0.03
Missed Flight	32	0.01
Property Loss	17	0.01
Property Damage/Personal Injury	14	0.01
Passenger Property Loss/Personal Injury	13	0.01
Passenger Property Loss/Personal Injur	8	0.00
Wrongful Death	4	0.00
Compliment	3	0.00
Not Provided	2	0.00
Bus Terminal	1	0.00

3. Average Payout per Year

Payout amounts remained relatively stable, with minor variation across years. Spikes in certain years may reflect policy shifts or major incidents.

The MEANS Procedure

Analysis Variable : Close_Amount					
Year	N Obs	Mean	Median	Maximum	Minimum
1996	1	50.0000000	50.0000000	50.0000000	50.0000000
2000	14	20.0458333	12.6100000	90.0000000	0
2001	15	34.2213333	0	195.3600000	0
2002	2158	257.8840093	59.0000000	25000.00	0
2003	24508	159.5073213	46.2550000	250000.00	0
2004	28960	132.8907268	37.5000000	125000.00	0
2005	22971	82.7919398	0	25000.00	0
2006	18806	79.2598353	0	12000.00	0
2007	18275	56.1089881	0	20000.00	0
2008	15902	56.9463276	0	16640.00	0
2009	11179	54.8347309	0	13000.00	0
2010	12215	63.7094733	0	10000.00	0
2011	10995	90.8000873	0	25000.00	0
2012	9997	89.3594084	0	6000.00	0
2013	9536	97.5646788	0	25483.44	0
2014	8680	97.2346554	0	5403.46	0
2015	7721	86.7086884	0	5000.00	0
2016	8188	72.5098259	0	2862.53	0
2017	8403	112.7732353	9.9900000	30000.00	0
2018	2	.	.	.	.

4. Claims by U.S. State

Geographic analysis reveals higher volumes in hub cities and coastal states. This may correlate with airport traffic density and urban claim likelihood.

The FREQ Procedure

StateName	Frequency	Percent
FLORIDA	27966	13.37
CALIFORNIA	25306	12.10
NEW YORK	17800	8.51
TEXAS	14378	6.87
ILLINOIS	9468	4.53
NEW JERSEY	7654	3.66
GEORGIA	7234	3.46
WASHINGTON	6953	3.32
PENNSYLVANIA	6928	3.31
ARIZONA	6494	3.10
NEVADA	6469	3.09
DIST. OF COLUMBIA	6033	2.88
COLORADO	5457	2.61
MASSACHUSETTS	4601	2.20
HAWAII	4425	2.12
MICHIGAN	4088	1.95
NORTH CAROLINA	4082	1.95
MINNESOTA	3601	1.72

5. Raw Data Snapshot

The initial dataset contained inconsistencies and missing data, especially in date and location fields.

*Program 1

CODE LOG RESULTS OUTPUT DATA

Table: PG1.TSA_CLAIMS View: Column names Filter: (none)

Columns

- Select all
- Claim_Number
- Date_Received
- Incident_Date
- Airport_Code
- Airport_Name
- Claim_Type
- Claim_Site
- Item_Category
- Close_Amount
- Disposition
- StateName
- State
- County
- City

Property	Value
Label	
Name	
Length	
Type	
Format	
Informat	

Total rows: 220855 Total columns: 14 Rows 1-100

	Claim_Number	Date_Received	Incident_Date	Airport_Code	Airport_Name	Claim_Type	Claim_S
1	2006081611123	17027	17006			Passenger Property Loss	Checke
2	2006062108380	16972	16957			Passenger Property Loss	Checkp
3	2006062008258	16972	16938			Passenger Property Loss	Checke
4	2006010699056	16831	16793			Passenger Property Loss	Checke
5	2006032303625	16880	16861			Property Damage	Checke
6	2006081511005	17021	.			Property Damage	Checke
7	2006082111318	17029	17015			Property Damage	Checke
8	2006070709101	16988	16979			Passenger Property Loss	Checkp
9	2006072109924	17003	16996			Passenger Property Loss	Checkp
10	2006041704637	16898	.			Passenger Property Loss	Checke
11	2006081010689	17016	16866			Property Damage	Checke
12	2006042505214	16905	16896			Property Damage	Checkp
13	2006061408002	16966	.				
14	2006081110819	17022	17014			Passenger Property Loss	Checke
15	2006051806567	16937	16924			Property Damage	Checke
16	2006040504066	16877	16855			Passenger Property Loss	Checke
17	2006073110251	17008	17001			Passenger Property Loss	Checke
18	2006042505160	16909	16863			Passenger Property Loss	Checkp
19	2006020801205	16834	16828			Passenger Property Loss	Checke
20	2006030302477	16859	16829			Passenger Property Loss	Checkp
21	2006032303589	16880	.			Passenger Property Loss	Checke
22	2006030702656	16861	16794			Property Damage	Checkp
23	2006042405125	16909	16889			Property Damage	Checke
24	2006061407915	16965	16941			Passenger Property Loss	Checke
25	2006062608528	16975	16948			Property Damage	Checke
26	2006050905781	16919	16899			Property Damage	Checke
27	2006020200854	16828	16800				
28	2006020701013	16832	16789			Property Damage	Checke
29	2006090611995	17021	.			Passenger Property Loss	Checke
30	2006021301461	16839	16821			Property Damage	Checkp
31	2006051906611	16937	16935			Passenger Property Loss	Checke
32	2006041904786	16904	16902			Passenger Property Loss	Checke
33	2006051506283	16931				Property Damage	Checke

6. Cleaned Data Preview

After cleaning, the dataset was trimmed to relevant rows with all required variables standardized.

Obs	Claim_Number	Date_Received	Incident_Date	Airport_Code	Airport_Name	Claim_Type	Claim_Site	Item_Category	Close_Amount	Disposition	StateName	State	County	City	Date_Issue
1	2006081611123	.	.			Passenger Property Loss	Checked Baggage	Candles - Decorative and other; Clothing - Shoes, belts, accessories, etc.; Dishes, Pottery, Glassware, Plasticware	.	Unknown					0
2	2006062108380	.	.			Passenger Property Loss	Checkpoint	Jewelry - Fine	.	Unknown					0
3	2006062008258	.	.			Passenger Property Loss	Checked Baggage	Cosmetics - Perfume, toilet articles, medicines, soaps, etc.; Medicines	.	Unknown					0
4	2006010699056	.	.			Passenger Property Loss	Checked Baggage	Other	.	Unknown					0
5	2006032303625	.	.			Property Damage	Checked Baggage	Luggage (all types including footlockers)	.	Unknown					0
6	2006081511005	.	.			Property Damage	Checked Baggage	Locks; Luggage (all types including footlockers)	.	Unknown					0
7	2006082111318	.	.			Property Damage	Checked Baggage	Locks	.	Unknown					0
8	2006070709101	.	.			Passenger Property Loss	Checkpoint	Jewelry - Fine	.	Unknown					0
9	2006072109924	.	.			Passenger Property Loss	Checkpoint	DVD/CD Players; Other	.	Unknown					0
10	2006041704637	.	.			Passenger Property Loss	Checked Baggage	Eyeglasses - (including contact lenses)	.	Unknown					0

Conclusion

This TSA claims case study demonstrates proficiency in data cleaning, exploratory analysis, and visual storytelling using SAS. The findings can be used to inform future risk mitigation strategies and improve claims processing pipelines.